

WEEK 1(Day 2)

1. Type Conversion

Type conversion means converting one data type into another data type.

(a) Integer to Float : Converts an integer into a decimal number

```
a = 10
b = float(a)

print(b)
print(type(b))

10.0
<class 'float'>
```

(b) Float to Integer : Converts a float into an integer by removing the decimal part.

```
a = 12.8
b = int(a)

print(b)
print(type(b))

12
<class 'int'>
```

c) Integer to String: Converts an integer into a string.

```
a = 25
b = str(a)

print(b)
print(type(b))

25
<class 'str'>
```

(d) String to Integer: Converts a numeric string into an integer.

```
a = "100"
b = int(a)

print(b)
print(type(b))

100
<class 'int'>
```

(e) Float to String :Converts a float into a string.

```
a = 45.6
b = str(a)

print(b)
print(type(b))

45.6
<class 'str'>
```

(f) String to Float:Converts a numeric string into a float.

```
a = "56.8"
b = float(a)

print(b)
print(type(b))

56.8
<class 'float'>
```

(g) Integer to Boolean:0 becomes False, any non-zero value becomes True.

```
a = 5
b = bool(a)

print(b)

True
```

(h) Boolean to Integer:True becomes 1, False becomes 0.

```
a = True
b = int(a)

print(b)
```

```
1
```

2.Types of Operators

(a) Arithmetic Operators:Used to perform mathematical calculations.

```
a = 10
b = 3

print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a % b)
print(a ** b)
print(a // b)

13
7
30
3.3333333333333335
1
1000
3
```

(b) Comparison Operators: Used to compare two values and return True or False.

```
a = 10
b = 5

print(a > b)
print(a < b)
print(a == b)
print(a != b)
print(a >= b)
print(a <= b)
```

```
True
False
False
True
True
False
```

(c) Logical Operators: Used to combine multiple conditions.

```
a = True
b = False

print(a and b)
print(a or b)
print(not a)
```

```
False
True
False
```

(d) Assignment Operators: Used to assign and update values of variables.

```
x = 10

x += 5
print(x)

x *= 2
print(x)
```

```
15
30
```

(e) Identity Operators: Used to check whether two variables refer to the same object.

```
a = [1, 2]
b = a

print(a is b)
print(a is not b)
```

True
False

(f) Membership Operators: Used to check whether a value exists in a sequence.

```
text = "Python"

print("P" in text)
print("z" not in text)
```

True
True

(g) Bitwise Operators: Used to perform operations on binary (bit) values.

```
a = 5
b = 3

print(a & b)
print(a | b)
print(a ^ b)
print(~a)
print(a << 1)
print(a >> 1)
```

1
7
6
-6
10
2

